One bulletin. Two reading experiences.

A faithful print edition, a practical phone editor, and a readable weekly digital archive.

세인트루이스 한인장로교회

Korean Presbyterian Church of St. Louis

4 pages
US Letter portrait

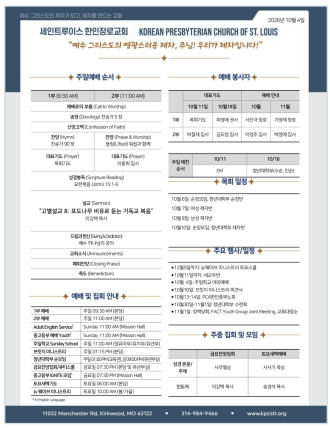
15 items
Numbered announcements

2 views
Print + mobile

The assignment

Replace the weekly Publisher workflow with a web application. The pastor can duplicate a prior week, enter content, ask AI for changes, edit directly, review the exact print layout, and export a PDF. Elders can review a specific revision. The finalized bulletin also becomes an accessible digital page and a dated archive entry.

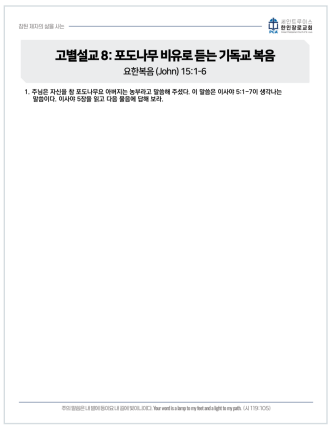
Mobile authoring is part of the core scope: the pastor must be able to make every routine content edit and propose revisions from a phone, not just read a published bulletin.



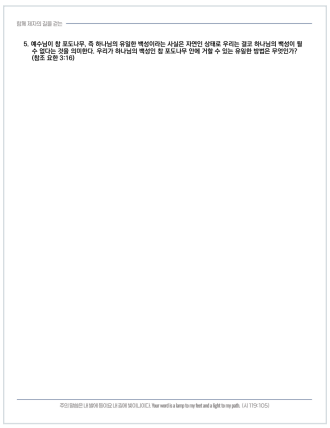
Worship



Announcements



Worksheet A



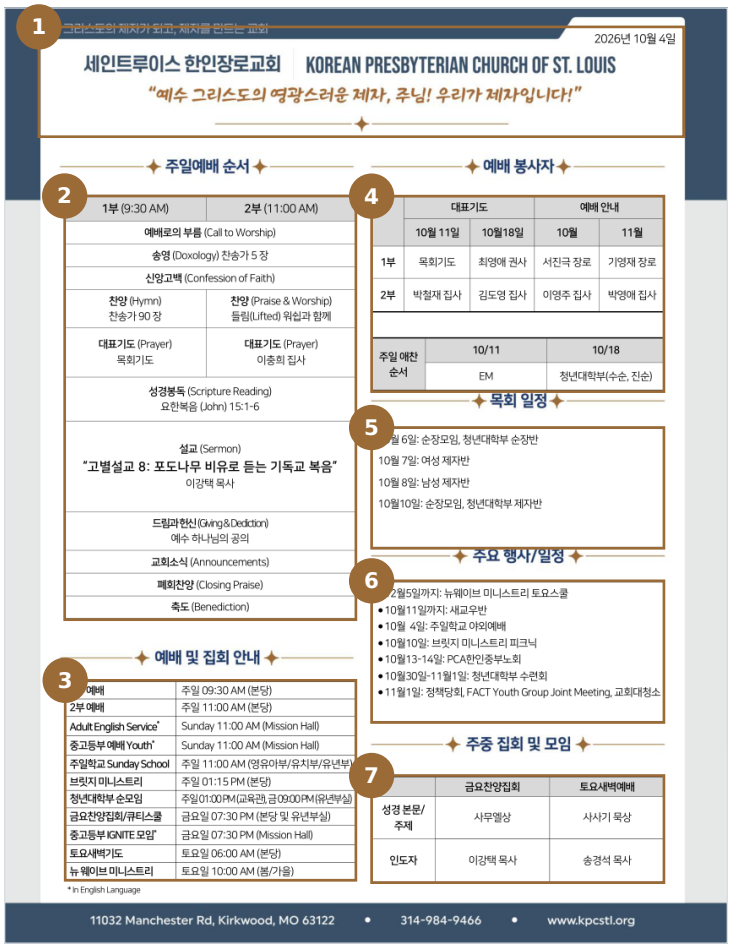
Worksheet B

The key design decision

Keep the page design fixed. Store the content as structured records. Use AI to propose field edits; use a deterministic layout engine to decide whether they fit.

Worship and weekly schedules

Measured from the supplied October 4, 2026 PDF. Coordinates use points from the top-left corner.



1 Identity and date

Church name, annual theme, ornamental rules, date tab, and address footer.

2 Sunday worship

11 order-of-service items. Shared rows span both services; hymn and prayer rows differ by service.

3 Regular service guide

11 schedule rows. Times, language notes, and locations belong to church settings.

4 Volunteers and meals

Two upcoming prayer dates, two usher months, and two meal-serving dates.

5-7 Weekly planning

Four pastoral entries, seven upcoming event bullets, and two midweek worship columns.

Region	x / y	Width / height
Worship order	50.83 / 159.20	246.62 / 358.83
Regular service guide	51.27 / 569.36	246.17 / 169.32
Volunteer roster	310.76 / 159.33	247.56 / 98.91
Meals	310.76 / 279.20	247.55 / 45.68
Pastoral / events	310.76 / 355.28; 485.11	247.55 / 102.26; 119.95
Midweek meetings	310.76 / 649.98	247.56 / 90.75

Main content columns are about 247 pt wide with a 13.3 pt gap. The page frame and section-heading positions must stay fixed while the contents change.

Announcements, people, and giving

This is the tightest page and the primary overflow risk.



1 Announcement panel

15 numbered items in two columns. Text begins at x=50.04 and x=231.05. Each item needs a title, body, and order.

2 Staff and giving sidebar

Three ministry groups, 11 role entries, and online offering instructions. One person has two roles.

3 Prayer-request QR

A separate bitmap near the lower-right of the announcement panel. Reserve this space during layout.

4 Prayer requests

Three bullets, including personal health intentions. Digital visibility should be chosen deliberately.

5 Offering statistics

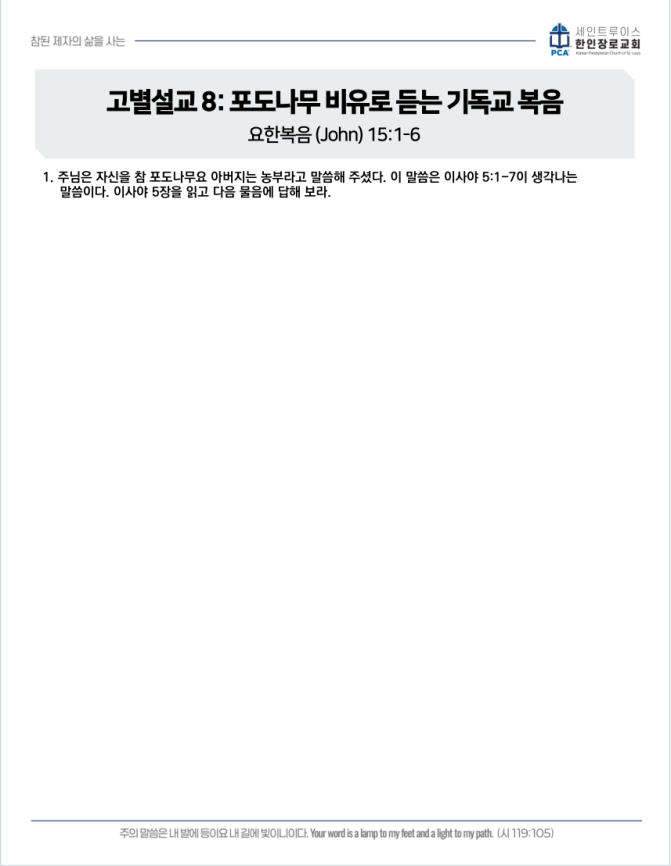
Seven giving categories and a computed total. The sample adds correctly to $21,287.40.

Area	Observed footprint / content behavior
Announcement text	First column x=50.04; second x=231.05; top text y=80.21. Approximate usable widths 175 / 173 pt. These are starting layout budgets, not a guarantee for every string.
Staff sidebar	Approximate frame x=424-569, y=38-627. Contact rows and bilingual role labels use mixed, small type.
Lower panels	Prayer region 247.55 x 107.63 pt. Giving region 249.15 x 93.45 pt plus a separate note.

The gray announcement area, sidebar frame, main headings, Zelle label, and offering QR are part of the full-page background image. They need their own template assets or an editable reconstruction.

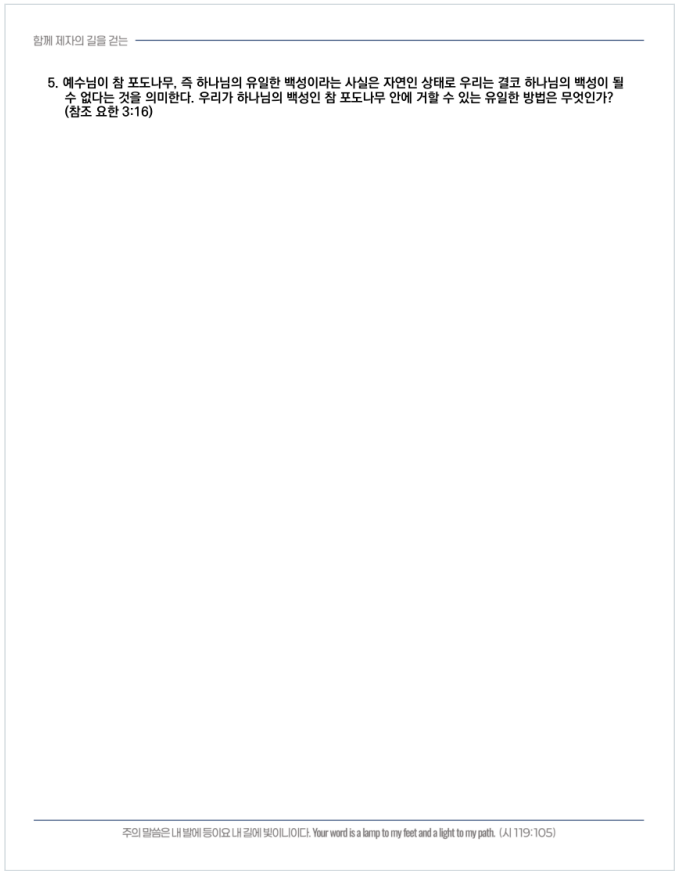
The sermon worksheets are a separate template

Preserve the large writing areas; do not automatically fill empty space with generated material.



PAGE 3 / WORKSHEET OPENING

24 pt sermon title, 17.04 pt passage, and 12 pt question text. Title and passage are live text over a light-gray banner image. Question 1 begins near x=38.88, y=153.98.



PAGE 4 / CONTINUATION

12 pt question text begins near x=38.88, y=63.09. The continuation header differs from page 3. Both pages include a small footer and a large blank response area.

Source gap to confirm

The supplied PDF visibly contains question 1 and question 5 only. Questions 2-4 are absent in both the rendered pages and extracted text. Preserve the numbering until the pastor supplies or confirms the intended questions.

Implementation: store questions by stable ID, display number, prompt, page placement, and answer-space allocation. A numbering-gap warning should invite correction; it must not invent questions or silently renumber this source.

How to preserve the existing layout

Exact matching is a validation target. It is not something an AI prompt alone can guarantee.

Property	Observed in the source	Implementation requirement
Paper	612 x 792 pt; 8.5 x 11 in	Lock to Letter portrait. Equivalent CSS canvas: 816 x 1056 px at 96 CSS px/in.
Primary type	NanumSquare R / B / EB	Obtain full licensed font files. The PDF embeds subsets; new Korean names may require missing glyphs.
Bulletin text	Mostly 9 pt; headings / labels vary	Match line height, weight, cell padding, and wrapping. Set a legibility floor for each region.
Small sidebar text	About 6.96-8.52 pt	Preserve the print reference; enlarge substantially in the mobile reader.
Worksheet type	KoPubWorldDotum Bold, 12 pt	Use the correct font metrics, not an approximate system fallback.
Other type	Bebas Neue, Calibri, Gill Sans, Batang, Symbol, Malgun Gothic	Some are limited to labels or special characters. Inventory which require editable equivalents.
Raster artwork	One full-page background per page	Reuse for the first fidelity pass. Backgrounds are roughly 150 dpi; this does not create higher-resolution artwork.

SOURCE-DERIVED PALETTE

#3A5068 #9B6B37 #EBECEE #D9D9D9

Colors are measured from embedded artwork and vector objects; JPEG artwork contains nearby color variations.

Two-stage template reconstruction

First, retain the source artwork and reproduce editable text and tables at measured positions. Then replace baked-in text and artwork with maintainable assets where needed, checking each change against the source.

Give every weekly value a home

Both the print renderer and mobile views read the same structured bulletin revision.

Content group	Stored fields	Update pattern
Issue metadata	Church ID; service date; revision; template version; status; language	Weekly / system
Worship order	Item type; Korean/English labels; service-specific value; hymn; prayer leader	Weekly + defaults
Sermon	Title; scripture reference; preacher; questions with number and answer space	Weekly
Volunteer roster	Service; role; person; assignment date or month	Weekly / monthly
Meals and events	Team; date; time; venue; description; start and end dates	Weekly
Announcements	Stable ID; title; rich text; order; start/expiry dates; visibility	Weekly / recurring
Prayer intentions	Text; effective week; audience; optional expiry	Weekly
Offering report	Category amounts as integer cents; period label; calculated total	Weekly
Church settings	Service times; contacts; roles; locations; theme; giving and prayer links	Occasional
Review and files	Comments; proposals; approval revision; PDF hash; published snapshot	Each revision

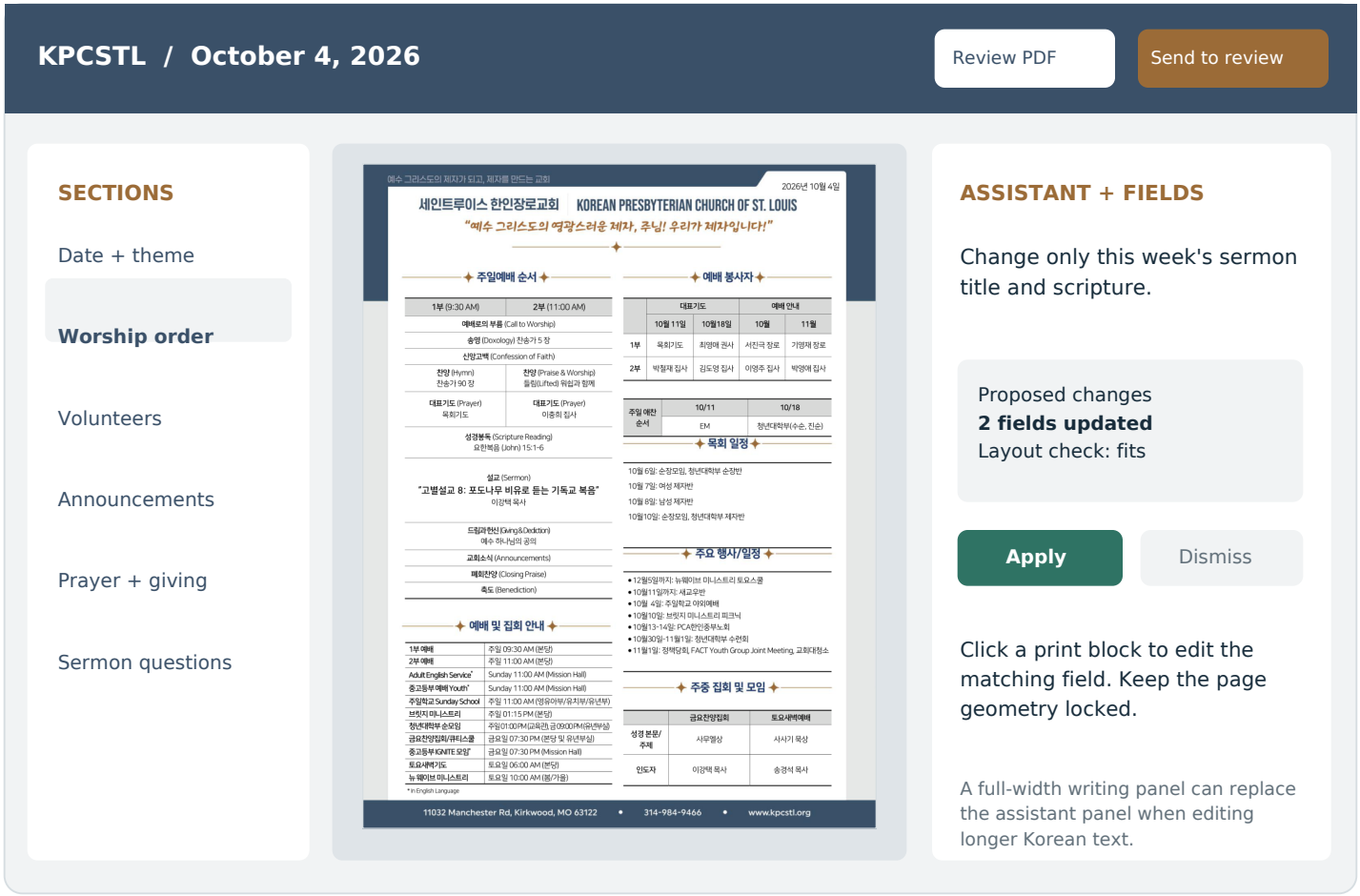
Weekly duplication must be selective

Carry forward recurring schedules and unexpired notices. Ask for confirmation of new prayer leaders, dates, giving amounts, sermon details, and one-time events. Never blindly roll every old value into the new week.

Retain original wording during migration. Each imported field should point back to its source page and region until a person confirms it. Repeated sermon information should reference one sermon record, so changing the title updates every view.

A calm workspace around the bulletin

The diagram below is a proposed interface, not a running application.



Direct editing without layout drift

Click a text region or select its section. Update a bound field with safe formatting. Preserve the live caret and Korean IME composition; avoid rebuilding the editor DOM on every keystroke.

Clear save and validation state

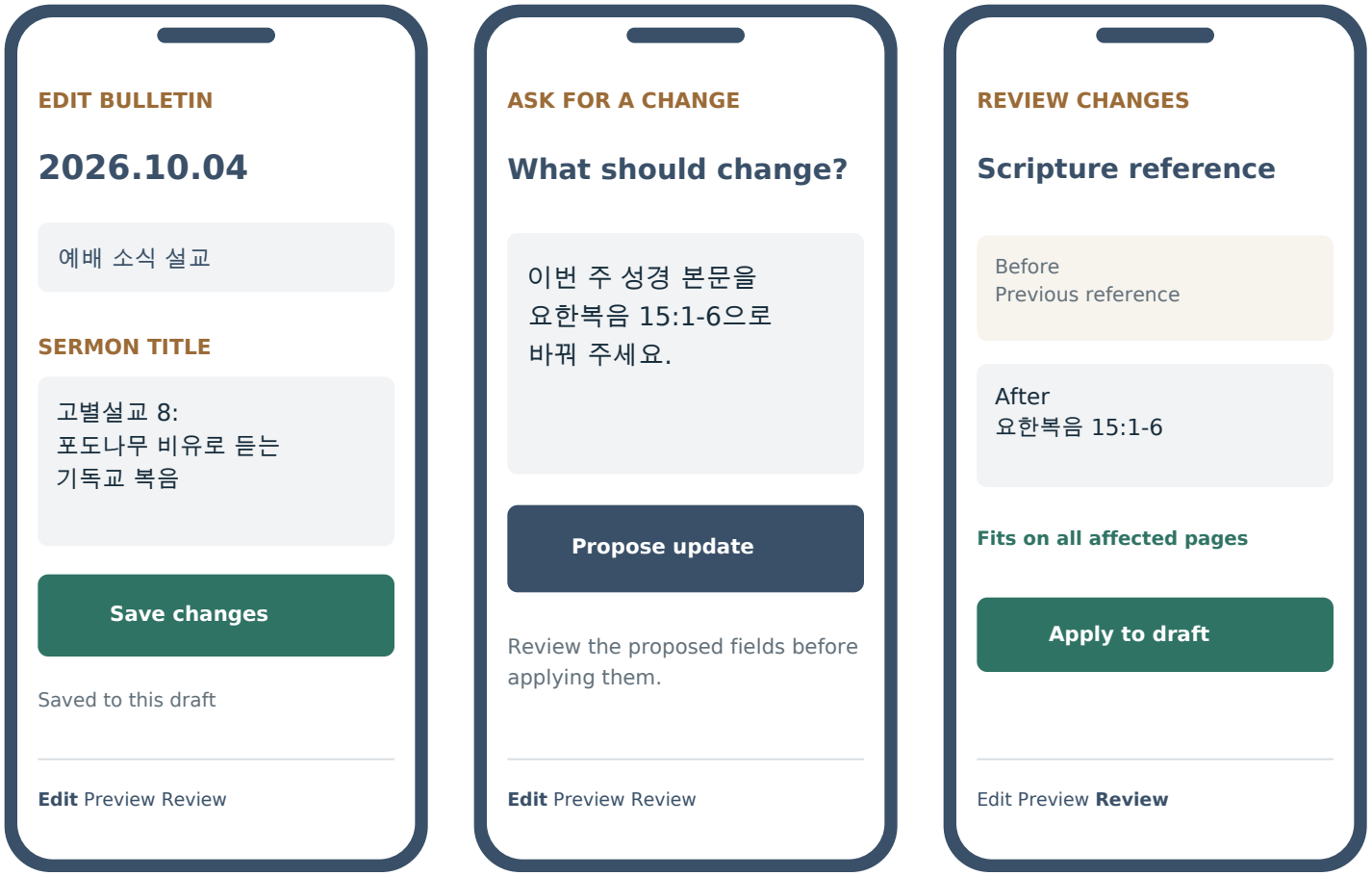
Show Saving, Saved, Offline, or Save failed. Highlight only the region that overflows. Provide undo, revision history, and a comparison against the last reviewed version.

One preview source

Screen print preview and PDF export use the same template definition. Page zoom changes viewing scale, never the actual font size or paper geometry.

The pastor can finish the work on his phone

Every routine content edit and proposed change must work on mobile, including review preparation.



All content tools, adapted to touch

Edit titles, names, dates, rosters, notices, giving values, and sermon questions. Add, delete, and reorder with accessible buttons; dragging is optional. Long text opens a full-screen editor.

Print preview stays available

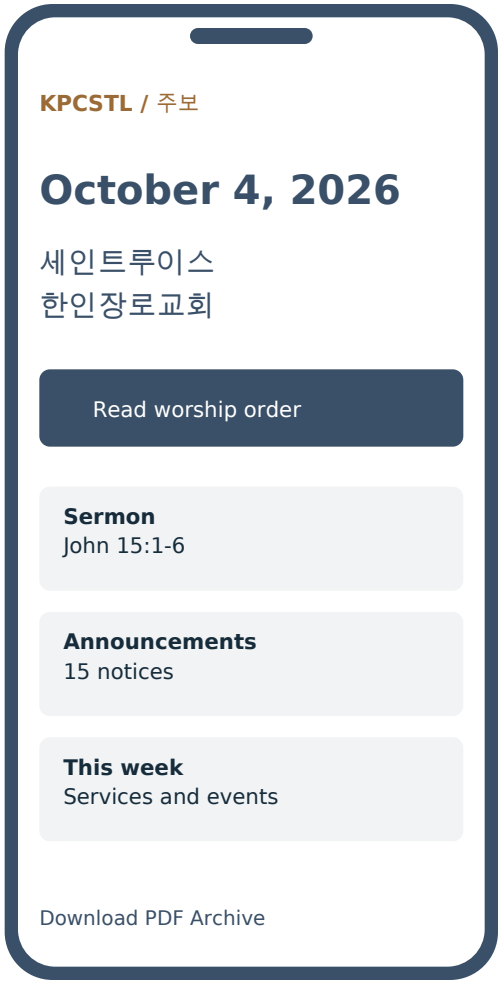
Switch to a page preview, pinch or tap to zoom, and tap a highlighted region to edit it. Offer PDF download and the device share sheet where supported.

Proposals are not lost

Draft changes save to the server. A proposal records its base revision and author. Conflicting edits show a comparison instead of overwriting another person's work. Offline changes wait for reconnect.

A digital bulletin people can comfortably read

Reflow the approved content for phones while preserving the church identity and section order.



Readable, not a miniature PDF

Use Korean text at a proposed 16-18 CSS px base size, comfortable line spacing, and touch targets around 44 px. Present rosters as cards or service tabs. Preserve the original PDF as a separate download.

A stable home and a dated issue

Use a permanent latest-bulletin link plus a unique URL for every service date. A QR code can point to the latest link, so the destination advances each week without reprinting the code.

Permanent weekly history

Every Sunday keeps its own dated entry, PDF, and mobile edition. Creating the next week never overwrites the past. Within each week, retain corrected revisions and a change log. Browse year/month/week and search title, scripture, or notices. Older PDFs can be imported by date; a mobile transcription needs review. A server-backed archive works across devices.

Proposed path or label	Purpose
/bulletins/latest	Latest published issue; never a draft
/bulletins/2026/10/04	Permanent digital edition for this date
Archive > 2026 > October > October 4	Folder-like navigation for people
2026/10/2026-10-04/r003/bulletin.pdf	Immutable file path for a specific revision

Paths above are design proposals, not live links. Public, member-only, and elder-review views may require different visibility rules. Personal prayer details and contacts should not become public merely because a mobile page exists.

AI proposes content. The renderer checks fit.

Use GPT-6 Astra through a server-side integration with reviewable, field-level changes. [W1, W2]

Step	Required behavior
1 Interpret	Accept Korean or English instructions plus the selected section and current revision. Send only the content needed for the requested change.
2 Propose	Return a strict structured proposal: base revision, field paths, old and new values, explanation, and unresolved questions. Do not return arbitrary executable HTML.
3 Validate	Check allowed fields, required values, date/day consistency, totals, numbering, and referential links. Schema conformity does not prove factual correctness.
4 Measure	Load the actual fonts and assets, render at the fixed print dimensions, and measure wrapping, overflow, clipping, and collisions in every affected region.
5 Review	Show a before/after diff and affected-page preview. Apply only the changes selected by the pastor. Keep undo and an audit record.
6 Export	Generate a PDF from that exact validated revision; record its content hash and template version.

A fixed four-page design cannot fit unlimited text

Start with the existing font sizes and spacing. If a section overflows, identify it and offer a shorter draft or an explicit layout choice. Never silently remove information, shrink it without limits, or create extra pages.

The fit policy

Keep the reference geometry locked by default. Let the pastor approve wording edits, limited spacing adjustments, or an additional insert if the church allows it. Store any exception so it is visible to reviewers.

No stalled spinner

Show AI job state, elapsed time, cancel, and a retry option. Persist the request and proposal separately from the saved draft. A timeout must not erase manual edits or falsely report success.

The official model identifier is gpt-6-astra. API access, billing, and credentials still need configuration for the church deployment. This study verifies documented support; it does not claim a live API call has been tested.

Approval belongs to a specific revision

The shared review PDF, phone review, and final publication must identify the same content snapshot.

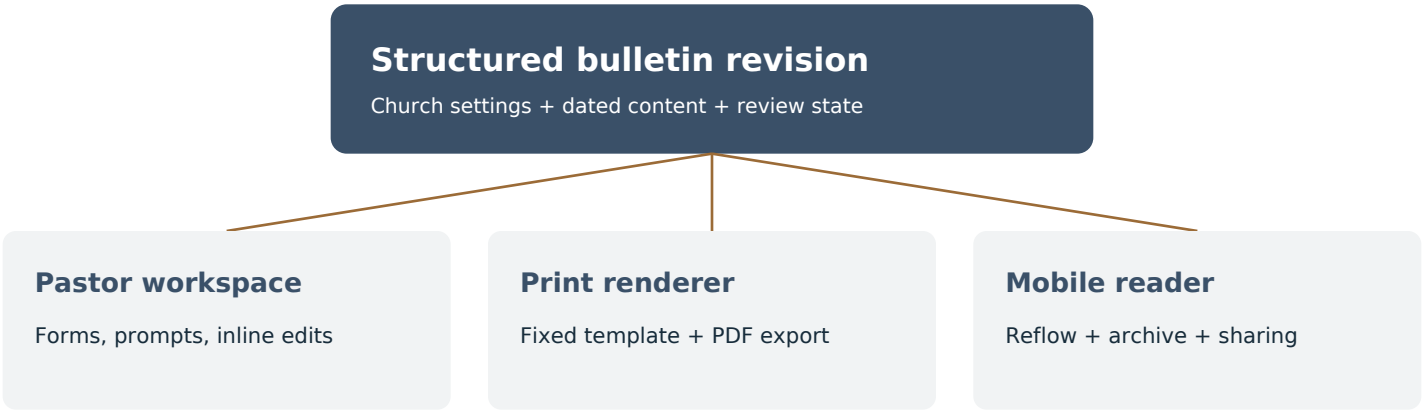
1. **Draft** — Pastor edits and requests proposals.
2. **In review** — Freeze revision for elder comments.
3. **Approved** — Record the approver and revision.
4. **Published** — Publish the matching PDF and digital view.

Role	Proposed permissions
Pastor / editor	Create weeks; edit every section on desktop or phone; apply proposals; export review PDFs.
Elder / reviewer	Read a specific revision; comment; propose replacements; approve if assigned.
Publisher / administrator	Finalize approved content; manage audience and template settings; correct publication via a new revision.
Congregation	Read and download published issues permitted for their audience.

Editing an approved or reviewed revision creates a new revision and invalidates that approval for the changed content. Keep old PDFs immutable; public links resolve only to an explicitly published snapshot. Saturday finalization is a deliberate action by default, not an automatic scheduled publish.

One content source, separate presentation layers

A proposed implementation structure; technology choices remain adaptable to the church hosting environment.



Layer	Proposed responsibility
Browser UI	TypeScript component UI; Korean-first labels; desktop canvas plus mobile Edit / Preview / Review. Accessible forms and shared validation.
Application server	Authentication, permissions, saving, proposal generation, version conflicts, comments, and publish transactions. API keys remain on the server.
Data store	Relational records for issues, immutable revisions, users, roles, comments, approvals, and publication pointers.
File storage	Original artwork, complete fonts, exported PDFs, and thumbnails. Separate private review files from publishable files.
PDF job service	Pinned Chromium version with the same print HTML/CSS as the preview; load fonts before export. Preserve backgrounds and CSS page size. [W3, W4]
AI service	Responses API with gpt-6-astra and structured proposals; explicit failures, cancellation, usage records, and bounded retries. [W1, W2]

The public GitHub repository contained only README.md at initial inspection. The project owner subsequently authorized including the already-public source bulletin as a reference fixture. Keep credentials and changing draft/review records in application services. Browser-local storage alone cannot provide the shared archive.

Prove fidelity before adding weekly automation

Each milestone has a clear pass condition rather than an unsupported promise of a perfect match.

Milestone	Deliverable	Pass condition
1 Reproduce	October 4 reference in an editable template	Four Letter pages; preserved geometry; correct fonts and line breaks; source comparison and church sign-off.
2 Manual workflow	Desktop and mobile content editors; save; revision history; PDF export	Every section can be edited on a phone; saved changes survive reload and appear correctly on another device.
3 AI proposals	Korean/English prompts; field diff; fit warnings; undo	Approved changes touch only intended fields; bad, stale, or overflowing proposals cannot silently publish.
4 Review + archive	Elder comments; revision approval; final publish; mobile issues and archive	Drafts stay private; review points to a fixed revision; final PDF and digital edition match.
5 Weekly rehearsal	Pastor and elders complete one real bulletin cycle	A new week can be prepared, reviewed, corrected, and finalized without Publisher.

Critical acceptance tests

Typography — New Korean names, mixed English/Korean text, long titles, and pasted line breaks preserve legibility.

Fit — A long announcement triggers a visible warning before export; no clipped text, accidental fifth page, or unapproved truncation.

Revision safety — Two editors changing the same item produce a recoverable conflict. A stale AI proposal cannot overwrite a newer edit.

Publishing — Published PDF, digital content, revision ID, and template version agree. A correction creates a traceable new revision.

Weekly archive — Create weeks across a month and year boundary. Both stay findable, with independent PDFs; a correction preserves the earlier revision.

Mobile — Real iPhone Safari and Android Chrome checks: keyboard, Korean IME, reorder, preview, save, PDF download, and share.

Reference comparison should combine a rendered overlay with manual inspection at print size. Antialiasing can differ between engines; agree on a visual tolerance before calling the reconstruction an exact match.

What is confirmed, and what needs a choice

Prepared October 24, 2026. This deliverable is a source study and proposed app specification.

Confirmed scope

Preserve the existing print design; edit weekly content through prompts and direct controls; support all routine edits and proposals on the pastor's phone; export review/final PDFs; support elder review and Saturday finalization; publish an optimized digital edition and a year/month/week archive.

Decision	Why it matters / working default
Original assets and worksheets	Request the Publisher .pub file plus full fonts and logos if available. Confirm whether questions 2-4 are intentionally absent and whether four pages are standard.
Public bulletin content	The owner confirmed that the supplied bulletin is already public and authorized including its personal information. Drafts and review-only discussions remain a separate workflow.
Review responsibilities	Identify who edits, who may approve, and who publishes. Default to a pastor/editor with elders as reviewers and a deliberate final publish.
Print constraints	Confirm Letter paper, separate sheets versus duplex/folding, actual printer margins, and whether an overflow insert is acceptable.
Deployment and AI ownership	Choose church-owned hosting/domain, sign-in method, and API billing owner. Reuse existing church accounts when practical.

EVIDENCE AND METHOD

S1. Attached October 4, 2026 bulletin: all four pages rendered and inspected; text, fonts, image resources, vector boundaries, and metadata extracted. Creator/producer: Microsoft Publisher for Microsoft 365. No editable .pub source was supplied.
S2. GitHub repository listing and README inspected: github.com/dohyoungkims/KPCSTL-Jubo. Only README.md was present at inspection.
Source quality notes: worksheet numbering jumps from 1 to 5; the baptism notice says 10/9 (Sunday), but October 9, 2026 is Friday; some Korean labels have damaged text extraction despite correct visual appearance. These are review flags, not authorized content corrections.

OFFICIAL TECHNICAL REFERENCES / VERIFIED OCTOBER 24, 2026

W1. GPT-6 Astra model - https://developers.openai.com/api/docs/models/gpt-6-astra
W2. Structured Outputs - https://developers.openai.com/api/docs/guides/structured-outputs
W3. Playwright PDF API - https://playwright.dev/docs/api/class-page#page-pdf
W4. W3C CSS Paged Media (Working Draft) - https://www.w3.org/TR/css-page-3/

Simple interactions, grounded in real tasks

Design recommendations informed by primary usability and accessibility sources; user testing remains to be done.

Design choice	Reason and application
Visible status	Show the selected Sunday, saved state, revision, and review status. A failed save must be visible and recoverable. [U1]
Recognizable controls	Use familiar Korean church labels and ordinary fields; prompting is optional. Keep editing, preview, review, and archive easy to find. [U1]
Stable phone navigation	Use persistent destinations rather than mixing navigation with publish/delete actions. Long editing forms need room for the phone keyboard. [U2]
Large touch controls	Target at least 44 x 44 CSS px for custom controls. WCAG minimum target guidance is lower; this project deliberately uses a more comfortable target. [U3]
Visible keyboard focus	Sticky menus and footers should not conceal the active control. Test focus, zoom, and Korean input on real phones and desktop keyboards. [U4]
Recoverable changes	Offer undo, before/after comparison, and explicit fit errors. Keep prior weeks and corrected revisions available. [U1]

The first usability rehearsal

Have the pastor create a new Sunday, change a prayer leader, add and reorder a notice, recover a mistaken deletion, propose a correction, and find an older bulletin. Observe completion and confusion rather than assuming the interface is easy.

The interactive prototype demonstrates these flows with browser-local data. Real AI, shared accounts/storage, actual elder approval, public issue links, and edited print-PDF generation are separate implementation milestones. Domain tests passed; real-browser visual and interaction QA was unavailable in the authoring environment.

PRIMARY RESEARCH REFERENCES

U1. Nielsen Norman Group: Ten Usability Heuristics

U2. Apple Human Interface Guidelines: Tab Bars

U3. W3C: Target Size Enhanced

U4. W3C: Focus Not Obscured